PsiSAT Stage 3
Command & Data Handling Board

STM32F411RET6: Complete Wiring & Build Guide

Project: PsiSAT 1U CubeSat, gamma-ray-burst detector

For: Muhammad Maalik (project lead) and the PCB layout partner

Tool: EasyEDA, JLCPCB / LCSC parts

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Contents

1. What Stage 3 does 3
2. Power and boot (Step 1) 4
3. Programming header, SWD (Step 2) 5
4. Safety pull-down resistors (Steps 3 and 4) 6
5. Radio control lines (Steps 5 and 6) 6
6. Clock crystals (Step 7) 7
7. Data storage flash (Step 8) 8
8. I2C bus to the charger (Step 9) 9
9. Other header signals (Step 10) 9
10. Debug UART header (Step 11) 10
11. Full stack header map 11
12. Unused STM32 pins 12
13. Full parts list (BOM) 12
14. PCB layout guide (for the layout person) 13
15. Checklist before ordering 14
16. Open items and changes needed on other boards 14
17. Glossary 15
18. Sources 16

> Every table lists **one wire per row**. Items marked **OPEN** are not verified yet; items marked **FIXED** were corrected from an earlier version.

1. What Stage 3 does

Stage 3 is the satellite's **brain**. It is one small computer chip, a **microcontroller** (a chip that runs a program you load into it): the **STM32F411RET6** from ST, in a 64-pin square package called **LQFP64**. In this guide it is called **U1**.

Its jobs are:

- **Count and timestamp gamma-ray pulses** coming from Stage 2 (signal `PULSE_IN`).
- **Store data** on a flash memory chip (U2) on this board.
- **Talk to the battery charger chip** on Stage 4 (TI BQ25713, I2C address **0x6B**) over I2C.
- **Control the antenna burn wire** (signal `BURN_CTRL` to Stage 4).
- **Control the radio module**, which lives on Stage 6 (RFM96W or RFM98W, 433 MHz band), through the stack header. **Not** the RFM95W, which is an 868/915 MHz part.
- **Switch the SiPM high-voltage bias** on Stage 2 on and off, and read it back.
- **Read status signals**: antenna deployed, and charger OK.

The boards (stages) are stacked and connect to each other through two **2x20 stack headers** (a left one and a right one). A header is a row of metal pins; 2x20 means 2 rows of 20 pins, 40 pins total.

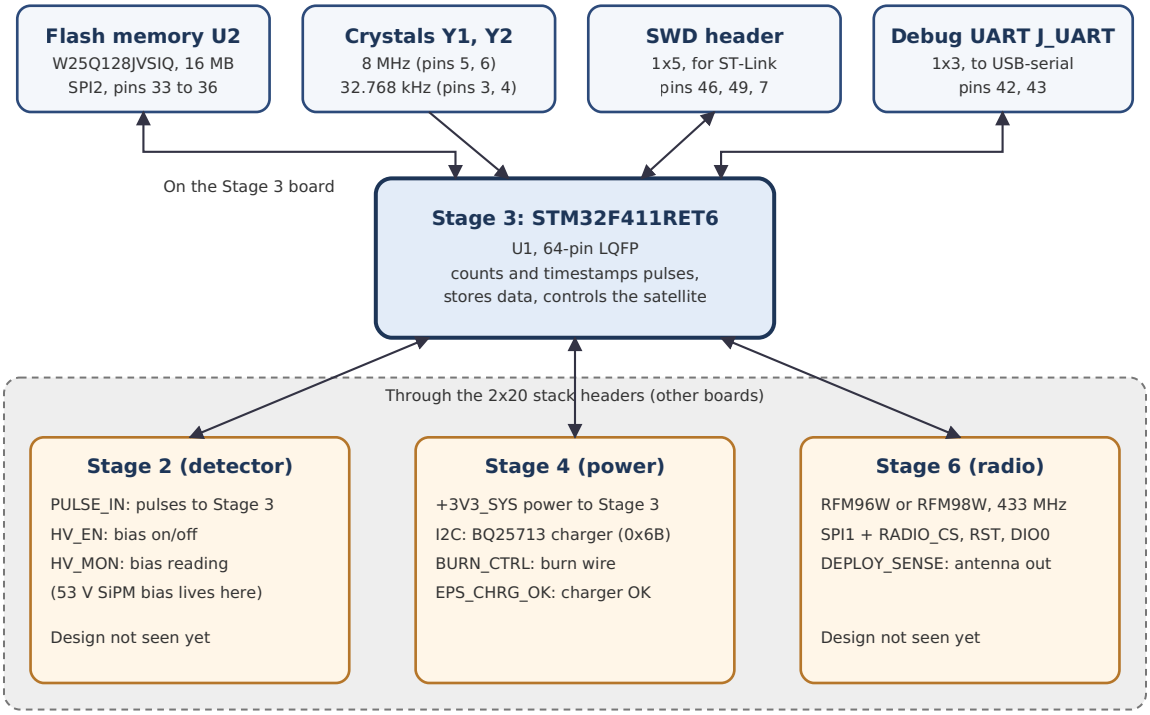


Figure 1. Stage 3 and what it connects to. Top row: parts on the Stage 3 board. Bottom row: other boards, reached through the stack headers.

1.1 How to read this guide

- **Net**: one electrical connection, that is, everything that is wired together. Every net has a name, like `+3V3_SYS` (the 3.3 V supply) or `GND` (ground, 0 V).
- **Net label**: a name tag in the schematic. Two wires with the same label are connected, even if no line is drawn between them. The spelling must match exactly.

- **"STM32 pin 14"** always means the physical pin number on the 64-pin chip U1. Names like PA0 or PB12 are the chip's internal names for that pin.
- **Pad 1 / pad 2**: the two metal ends of a small resistor or capacitor. For these parts it does not matter which end is which.
- **One wire per row.** Each table row is one connection. Tick it off in the schematic as you go.
- **Part sizes**: "0603" is a standard tiny surface-mount size (about 1.6 mm x 0.8 mm).
- Items marked OPEN are not yet verified. Items marked FIXED were wrong in an earlier version and are now corrected.

2. Power and boot (Step 1)

The chip gets its power from Stage 4 on the net `+3V3_SYS` (3.3 V), which comes in on **left header pins 3 and 4**. Ground is the net `GND`.

A **decoupling capacitor** is a small capacitor placed right next to a power pin; it acts like a tiny local battery that smooths out fast dips in the supply. A **bulk capacitor** is a bigger one that covers slower dips. ST requires these capacitors for the chip to start and run stably (ST DS10314; AN4488 section 2.2).

#	From	To	Part / note
1	STM32 pin 1 (VBAT)	+3V3_SYS (net label)	Backup supply. No separate battery.
2	100 nF cap (VBAT) pad 1	STM32 pin 1	100 nF, C14663
3	100 nF cap (VBAT) pad 2	GND	100 nF, C14663
4	STM32 pin 19 (VDD)	+3V3_SYS	Main supply pin
5	100 nF cap (pin 19) pad 1	STM32 pin 19	100 nF, C14663; place right at the pin
6	100 nF cap (pin 19) pad 2	GND	100 nF, C14663
7	STM32 pin 32 (VDD)	+3V3_SYS	Main supply pin
8	100 nF cap (pin 32) pad 1	STM32 pin 32	100 nF, C14663; place right at the pin
9	100 nF cap (pin 32) pad 2	GND	100 nF, C14663
10	4.7 µF bulk cap pad 1	+3V3_SYS (at pin 32)	4.7 µF, C19666; place at pin 32
11	4.7 µF bulk cap pad 2	GND	4.7 µF, C19666
12	STM32 pin 48 (VDD)	+3V3_SYS	Main supply pin
13	100 nF cap (pin 48) pad 1	STM32 pin 48	100 nF, C14663; place right at the pin
14	100 nF cap (pin 48) pad 2	GND	100 nF, C14663
15	STM32 pin 64 (VDD)	+3V3_SYS	Main supply pin
16	100 nF cap (pin 64) pad 1	STM32 pin 64	100 nF, C14663; place right at the pin
17	100 nF cap (pin 64) pad 2	GND	100 nF, C14663
18	STM32 pin 13 (VDDA)	+3V3_SYS	Analog supply
19	1 µF cap (VDDA) pad 1	STM32 pin 13	1 µF, C59782

#	From	To	Part / note
20	1 µF cap (VDDA) pad 2	GND	1 µF, C59782
21	100 nF cap (VDDA) pad 1	STM32 pin 13	100 nF, C14663
22	100 nF cap (VDDA) pad 2	GND	100 nF, C14663
23	STM32 pin 12 (VSSA)	GND	Analog ground
24	STM32 pin 18 (VSS)	GND	
25	STM32 pin 31 (VSS)	GND	**FIXED** was missing
26	STM32 pin 47 (VSS)	GND	
27	STM32 pin 63 (VSS)	GND	
28	STM32 pin 30 (VCAP_1)	4.7 µF cap (VCAP) pad 1	4.7 µF, C19666. **FIXED** was 2.2 µF
29	4.7 µF cap (VCAP) pad 2	GND	Place right at pin 30
30	STM32 pin 60 (BOOT0)	10 kΩ resistor (BOOT0) pad 1	10 kΩ, C98220. **FIXED** was on pin 50
31	10 kΩ resistor (BOOT0) pad 2	GND	Always boot own program
32	STM32 pin 7 (NRST)	100 nF cap (NRST) pad 1	Already on board
33	100 nF cap (NRST) pad 2	GND	Already on board

2.1 Why each item is there

- **VCAP_1 (pin 30)**: the chip makes its own internal core voltage and needs a **4.7 µF** capacitor here with **ESR below 1 Ω** (ESR is the small built-in resistance of a capacitor; normal ceramic capacitors easily meet this). Source: ST DS10314 Table 16. The 64-pin package has only **one** VCAP pin.
- **BOOT0 (pin 60)**: the 10 kΩ resistor to GND makes the chip always start its own program. Optional recovery trick: hold BOOT0 at 3.3 V while resetting to start ST's built-in bootloader (a small loader program that ST puts inside every chip).
- **NRST (pin 7)**: the reset pin. The chip has an internal pull-up, so no extra resistor is needed, only the 100 nF that is already there.

3. Programming header, SWD (Step 2)

SWD (Serial Wire Debug) is ST's 2-wire connection for loading the program into the chip and debugging it, using an **ST-Link** programmer. On this board it is a simple 1x5 pin header.

Part: 1x5, 2.54 mm pitch pin header, XFCN **PZ254V-11-05P**, LCSC **C492404**.

#	From	To	Part / note
1	SWD header pin 1	+3V3_SYS	Tells the ST-Link the board voltage
2	SWD header pin 2	STM32 pin 49 (PA14, SWCLK)	Clock
3	SWD header pin 3	GND	
4	SWD header pin 4	STM32 pin 46 (PA13, SWDIO)	Data
5	SWD header pin 5	STM32 pin 7 (NRST)	Reset

4. Safety pull-down resistors (Steps 3 and 4)

A **pull-down resistor** is a resistor from a signal to GND. It holds the signal at "off" (0 V) whenever nothing else is driving it. (A **pull-up** is the same idea but to 3.3 V, holding the signal "on".) This matters because **STM32 pins float** (are undefined) during reset and while the chip is being programmed.

Part for all three: 100 kΩ, 0603, 1%, UNI-ROYAL **0603WAF1003T5E**, LCSC **C25803**.

> **Part number correction:** an earlier list used LCSC **C106245**. That is wrong: it is a 10 pF capacitor. Use **C25803**.

#	From	To	Part / note
1	R_BURN pad 1	STM32 pin 14 (PA0, BURN_CTRL)	100 kΩ, C25803
2	R_BURN pad 2	GND	
3	STM32 pin 14 (BURN_CTRL)	Left header pin 13	Goes to Stage 4 burn MOSFET
4	R_PULSE pad 1	STM32 pin 15 (PA1, PULSE_IN)	100 kΩ, C25803
5	R_PULSE pad 2	GND	
6	STM32 pin 15 (PULSE_IN)	Left header pin 16	Comes from Stage 2

- **R_BURN** keeps the burn wire **OFF** while the chip is resetting or being programmed. Stage 4 also needs its **own** 100 k pull-down on the gate of its burn MOSFET (AO3400A).
- **R_PULSE** keeps the pulse input from picking up noise when Stage 2 is unplugged. Pin 15 is a **timer capture input** (TIM2_CH2): the chip's timer hardware records the exact time each pulse arrives.
- **R_HVEN** (the third 100 kΩ) is in section 9.

5. Radio control lines (Steps 5 and 6)

The radio itself is on **Stage 6**. Stage 3 only sends control signals to it through the left header. There is **no antenna and no radio power part on Stage 3**.

The radio uses **SPI**: a fast 4-wire link where the chip sends a clock (SCK), sends data out (MOSI), receives data back (MISO), and uses a chip-select line (CS) to say "I am talking to you now". Here it is the chip's SPI1 bus.

Left header pin	STM32 pin	Net
14	pin 21 (PA5)	`SPI_SCK`
15	pin 22 (PA6)	`SPI_MISO`
16	pin 15 (PA1)	`PULSE_IN` (not radio; listed for completeness)
17	pin 23 (PA7)	`SPI_MOSI`
18	pin 24 (PC4)	`RADIO_CS`
19	pin 16 (PA2)	`RADIO_RST`
20	pin 17 (PA3)	`RADIO_DIO0`

5.1 Radio chip-select pull-up

Part: R_CS (radio), 10 kΩ, YAGEO **RC0603FR-0710KL**, LCSC **C98220**.

#	From	To	Part / note
1	R_CS (radio) pad 1	RADIO_CS (STM32 pin 24)	10 kΩ, C98220
2	R_CS (radio) pad 2	+3V3_SYS	

This keeps the radio **deselected** (ignoring the bus) when Stage 6 is unplugged or the STM32 is resetting.

5.2 Rules

- **RADIO_RST and RADIO_DIO0: no resistors.**
- **Firmware rule:** RADIO_RST is only ever pulled **low** (for at least 100 µs) and then released. It is **never driven high** (Semtech SX1276 datasheet section 7.2).
- SPI1 clock: **10 MHz maximum** for the radio.
- PULSE_IN (header 16) sits between SPI_MISO (15) and SPI_MOSI (17). Firmware should avoid radio SPI traffic during a burst trigger.

> **Stage 6 needs (later):** RFM96W or RFM98W module, 100 nF + 10 µF at its supply pin, a u.FL antenna connector, and the same header pin numbers as above.

6. Clock crystals (Step 7)

These are small **quartz timing parts** that make the chip's clock accurate. They are **not** the CsI(Tl) gamma-ray detector crystal.

- **8 MHz crystal (Y1)** gives accurate timestamps. Without it, the chip's built-in clock can be off by up to about ±4%.
- **32.768 kHz crystal (Y2)** runs the real-time clock, so the 45-minute wait after deployment is right. Without it, "45 minutes" could really be about 31 to 79 minutes.

6.1 8 MHz crystal Y1

Y1: YXC **X32258MOB4SI**, 8 MHz, 12 pF, 3225 size, 4 pads, LCSC **C2682775**. **C_H1, C_H2:** 15 pF C0G 0603, LCSC **C1644** (C0G is a very stable capacitor type). **R_EXT:** 0 Ω 0603, LCSC **C21189**.

#	From	To	Part / note
1	Y1 pad 1	STM32 pin 5 (PH0-OSC_IN)	
2	Y1 pad 1	C_H1 pad 1	15 pF, C1644
3	Y1 pad 3	C_H2 pad 1	15 pF, C1644
4	Y1 pad 3	R_EXT pad 1	0 Ω, C21189
5	R_EXT pad 2	STM32 pin 6 (PH1-OSC_OUT)	
6	Y1 pad 2	GND	Case pad
7	Y1 pad 4	GND	Case pad
8	C_H1 pad 2	GND	
9	C_H2 pad 2	GND	

R_EXT is a **placeholder** OPEN: it may become up to 220 Ω if the first board shows the crystal is being driven too hard.

6.2 32.768 kHz crystal Y2

Y2: Epson **FC-135R**, 32.768 kHz, 7 pF, LCSC **C94674** (it can be placed either way round). **C_L1, C_L2:** 3.9 pF C0G 0603, LCSC **C1662**.

#	From	To	Part / note
10	Y2 pad 1	STM32 pin 3 (PC14-OSC32_IN)	
11	Y2 pad 1	C_L1 pad 1	3.9 pF, C1662
12	Y2 pad 2	STM32 pin 4 (PC15-OSC32_OUT)	
13	Y2 pad 2	C_L2 pad 1	3.9 pF, C1662
14	C_L1 pad 2	GND	
15	C_L2 pad 2	GND	

6.3 Notes

- **Why the 7 pF Epson part:** ST says the common 12.5 pF version will not start reliably on this chip (AN2867; AN4488 section 6.2).
- The capacitor values assume about 5 pF of stray capacitance from the board itself (ST's typical figure). They **may need fine-tuning on the first board** OPEN.
- **Firmware:** set `HSE_VALUE = 8000000`. Turn on LSE high-drive mode (`LSEMOD = 1`) **before** starting the 32.768 kHz oscillator (LSE).
- **Layout:** crystals and their capacitors within about 5 mm of pins 3 to 6; no traces running under them; keep pin 2 (PC13) unused and quiet; crystal grounds go to GND near pin 63.

7. Data storage flash (Step 8)

U2: Winbond **W25Q128JVSIQ**, 16 MB flash memory, **SOIC-8 208-mil (wide)** package. LCSC **C97521** (JLCPCB has stock; LCSC itself was out on Oct 7) or **C113767** (same chip, LCSC stock). It talks to the STM32 on the chip's second SPI bus, **SPI2**.

C_FL: 100 nF, LCSC **C14663**. **R_CS (flash), R_WP, R_IO3:** 10 kΩ, LCSC **C98220**.

#	From	To	Part / note
1	U2 pin 1 (/CS)	STM32 pin 33 (PB12)	Net `FLASH_CS`
2	U2 pin 2 (DO)	STM32 pin 35 (PB14)	Net `SPI2_MISO`
3	U2 pin 3 (/WP)	R_WP pad 1	10 kΩ, C98220
4	U2 pin 4 (GND)	GND	
5	U2 pin 5 (DI)	STM32 pin 36 (PB15)	Net `SPI2_MOSI`
6	U2 pin 6 (CLK)	STM32 pin 34 (PB13)	Net `SPI2_SCK`
7	U2 pin 7 (IO3)	R_IO3 pad 1	10 kΩ, C98220
8	U2 pin 8 (VCC)	+3V3_SYS	
9	C_FL pad 1	U2 pin 8	100 nF, C14663
10	C_FL pad 2	GND	
11	R_CS (flash) pad 1	FLASH_CS	10 kΩ, C98220
12	R_CS (flash) pad 2	+3V3_SYS	

#	From	To	Part / note
13	R_WP pad 2	+3V3_SYS	
14	R_IO3 pad 2	+3V3_SYS	

7.1 Notes

- On this exact chip, pin 7 is a plain data pin (IO3). It has no hold or reset function.
- Keep the SPI2 traces **shorter than 25 mm**. No series resistors are needed.
- **Capacity:** about 24 days of 1-second count totals, or about 70 minutes if every event is timestamped at 1,000 events per second. That is enough for counts plus short burst windows.
- **Radiation:** no radiation test or flight record was found for this chip family, and Winbond says it is not meant for space. Firmware should use checksums and keep 2 copies of critical data.
- **Firmware:** SPI mode 0; start at 12.5 MHz; reading the chip ID should return `EF 40 18`; PB12 is used as a plain on/off pin (GPIO) for chip select.

8. I2C bus to the charger (Step 9)

I2C is a 2-wire shared bus (clock SCL and data SDA) where each chip has its own address. Both wires need a **pull-up** resistor to 3.3 V to work.

R_SCL, R_SDA: 4.7 kΩ, 0603, 1%, YAGEO **RC0603FR-074K7L**, LCSC **C99782** (cheaper JLCPCB Basic alternative: **C23162**).

#	From	To	Part / note
1	STM32 pin 61 (PB8)	I2C_SCL	
2	STM32 pin 62 (PB9)	I2C_SDA	
3	R_SCL pad 1	I2C_SCL	4.7 kΩ, C99782
4	R_SCL pad 2	+3V3_SYS	
5	R_SDA pad 1	I2C_SDA	4.7 kΩ, C99782
6	R_SDA pad 2	+3V3_SYS	
7	Left header pin 12	I2C_SCL	To Stage 4
8	Left header pin 11	I2C_SDA	To Stage 4

- Keep this **one** set of pull-ups on Stage 3. Later, remove (do not fit) the duplicate 4.7 k pull-ups on Stage 4.
- Nothing added later may use I2C address **0x6B**; that is the BQ25713 charger.

9. Other header signals (Step 10)

R_HVEN: 100 kΩ, LCSC **C25803**.

#	From	To	Part / note
1	Left header pin 21	GND	Shield between radio lines and these signals
2	Left header pin 22	STM32 pin 29 (PB10)	Net `HV_EN`

#	From	To	Part / note
3	R_HVEN pad 1	HV_EN	100 kΩ, C25803
4	R_HVEN pad 2	GND	Keeps the bias OFF by default
5	Left header pin 23	STM32 pin 26 (PB0, ADC input 8)	Net `HV_MON`
6	Left header pin 24	STM32 pin 37 (PC6)	Net `DEPLOY_SENSE`
7	Left header pin 25	STM32 pin 38 (PC7)	Net `EPS_CHRG_OK`
8	Left header pin 26	GND	

9.1 What these signals mean

- **HV_EN** turns Stage 2's 53 V SiPM bias on and off. The pull-down keeps it **OFF** by default.
- **HV_MON** reads the bias voltage with the chip's **ADC** (a circuit that measures a voltage and turns it into a number). Stage 2 **must** divide the voltage down to 3.3 V or less first.
- **DEPLOY_SENSE** is the antenna-deployed switch from Stage 6. Whether it uses a pull-up or pull-down is pending the Stage 6 design **OPEN**.
- **EPS_CHRG_OK** is the "charger OK" signal from BQ25713 pin 4. Its 10 k pull-up lives on Stage 4.

10. Debug UART header (Step 11)

UART is a simple 2-wire serial link (one wire to transmit, TX, one to receive, RX), used here to print debug messages to a laptop. It is also a port for ST's built-in bootloader.

J_UART: 1x3, 2.54 mm pin header, XFCN **PZ254V-11-03P**, LCSC **C492402**.

#	From	To	Part / note
1	J_UART pin 1	STM32 pin 42 (PA9)	Net `UART_TX`
2	J_UART pin 2	STM32 pin 43 (PA10)	Net `UART_RX`
3	J_UART pin 3	GND	

- There is **no 3.3 V** on this header.
- Use a **3.3 V** USB-serial adapter and **cross** the wires: adapter TX to board RX, adapter RX to board TX.
- Firmware: USART1 on alternate function AF7.

11. Full stack header map

This is the **final** map. All stages must use these **same pin numbers**.

> The header map in section 4 of Stage3-CDH-Wiring-Review.md, and its radio-on-Stage-3 text in sections 3.3 and 3.4, are **outdated**. Use this table instead.

11.1 Left 2x20 header

"not used" means Stage 3 makes no connection to that pin.

Pin	Net	STM32 pin	Pin	Net	STM32 pin
1	`GND`		15	`SPI_MISO`	pin 22 (PA6)
2	`GND`		16	`PULSE_IN`	pin 15 (PA1)
3	`+3V3_SYS`	supply	17	`SPI_MOSI`	pin 23 (PA7)
4	`+3V3_SYS`	supply	18	`RADIO_CS`	pin 24 (PC4)
5	`+5V0_SYS`	not used	19	`RADIO_RST`	pin 16 (PA2)
6	`+5V0_SYS`	not used	20	`RADIO_DIO0`	pin 17 (PA3)
7	`VSYS`	not used	21	`GND`	
8	`VSYS`	not used	22	`HV_EN`	pin 29 (PB10)
9	`VBAT_CHG`	not used	23	`HV_MON`	pin 26 (PB0)
10	`VBAT_CHG`	not used	24	`DEPLOY_SENSE`	pin 37 (PC6)
11	`I2C_SDA`	pin 62 (PB9)	25	`EPS_CHRG_OK`	pin 38 (PC7)
12	`I2C_SCL`	pin 61 (PB8)	26	`GND`	
13	`BURN_CTRL`	pin 14 (PA0)	27 to 40	spare	not connected
14	`SPI_SCK`	pin 21 (PA5)			

11.2 Right 2x20 header (existing, from Stage 4)

Header pin	Net	Use on Stage 3
1, 2	`GND`	Connected to Stage 3 GND
3, 4	`V_SOLAR_POS`	Not connected on Stage 3
7	`BURN_OUT` (Stage 4 to Stage 6)	Not connected on Stage 3

Stage 3 connects **only GND** on the right header.

12. Unused STM32 pins

Leave these pins **unconnected** and put a **no-connect flag** (the small X symbol) on each one in the schematic, so the checker knows they are empty on purpose. Firmware sets them to analog mode (the lowest-power, quietest setting).

STM32 pin(s)	Name(s)
2	PC13
8 to 11	PC0 to PC3
20	PA4
25	PC5
27, 28	PB1, PB2
39 to 41	PC8, PC9, PA8
44, 45	PA11, PA12
50	PA15
51 to 54	PC10, PC11, PC12, PD2
55 to 59	PB3 to PB7

13. Full parts list (BOM)

BOM means "bill of materials": the shopping list. LCSC is the parts supplier used by the JLCPCB board factory.

Ref	Qty	Value / what	Mfr part number	LCSC
U1	1	Microcontroller, LQFP64 (already on board)	ST STM32F411RET6	
U2	1	16 MB SPI flash, SOIC-8 208-mil	Winbond W25Q128JVSIQ	C97521 or C113767
Y1	1	8 MHz crystal, 12 pF, 3225	YXC X32258MOB4SI	C2682775
Y2	1	32.768 kHz crystal, 7 pF	Epson FC-135R	C94674
VCAP, bulk	2	4.7 µF (pin 30 VCAP + bulk at pin 32)	Samsung CL10A475KO8NNNC	C19666
VDDA	1	1 µF (VDDA)	Samsung CL10B105KO8NNNC	C59782
Decoupling	8	100 nF (VDD 19/32/48/64, VDDA, VBAT, NRST, flash C_FL); NRST one already on board	YAGEO CC0603KRX7R9BB104	C14663
C_H1, C_H2	2	15 pF C0G 0603		C1644
C_L1, C_L2	2	3.9 pF C0G 0603		C1662
R_EXT	1	0 Ω 0603 (placeholder)		C21189
10 k	5	10 kΩ (BOOT0, RADIO_CS, FLASH_CS, flash WP, flash IO3)	YAGEO RC0603FR-0710KL	C98220 (Basic alt. C25804)

Ref	Qty	Value / what	Mfr part number	LCSC
R_SCL, R_SDA	2	4.7 kΩ	YAGEO RC0603FR-074K7L	C99782 (Basic alt. C23162)
R_BURN, R_PULSE, R_HVEN	3	100 kΩ (BURN_CTRL, PULSE_IN, HV_EN)	UNI-ROYAL 0603WAF1003T5E	C25803
SWD header	1	1x5 2.54 mm header	XFCN PZ254V-11-05P	C492404
J_UART	1	1x3 2.54 mm header	XFCN PZ254V-11-03P	C492402
Stack headers	2	2x20 stack headers (left, right), already in design; must be long-pin stacking type so pins pass through	OPEN NOT verified	OPEN NOT verified

> Stock and JLCPCB Basic/Extended status were checked Oct 7 to 8, 2026 and can change. Re-check before ordering.

14. PCB layout guide (for the layout person)

The board is set to **4 copper layers** in EasyEDA: Top, Inner1, Inner2, Bottom. A **pour** (or **plane**) is a large filled area of copper connected to one net. A **via** is a small plated hole that connects copper on different layers.

Layer	What goes on it
Top	Parts and signal traces
Inner1	Solid `GND` pour over the whole board
Inner2	Solid `+3V3_SYS` pour over the whole board
Bottom	Extra signal traces, plus a `GND` pour over the leftover space

14.1 Steps

a. On **Inner1**, draw a copper pour just inside the board outline and assign it to net `GND`.
b. On **Inner2**, do the same and assign it to net `+3V3_SYS`.
c. On **Bottom**, add a `GND` pour over the leftover space.
d. **Placement rules:**
   - Each 100 nF capacitor right at its VDD pin (19, 32, 48, 64).
   - VCAP 4.7 µF right at pin 30.
   - Crystals and their capacitors within about 5 mm of pins 3 to 6, with no traces under them.
   - Flash U2 close to pins 33 to 36 (SPI2 traces shorter than 25 mm).
   - Headers placed per the header map (section 11).
e. Use vias to connect the capacitor GND pads down to the Inner1 GND plane.
f. Run **DRC** (Design Rule Check: the software checks spacing, widths and holes against the factory's limits) using JLCPCB's rules.

> A 4-layer board costs more than a 2-layer board at JLCPCB. Check their online quote tool for the current price.

15. Checklist before ordering

ERC (Electrical Rule Check) is the schematic checker: it flags things like unconnected pins or two outputs tied together.

	Check	Status
☐	Every part has a unique name (U1, U2, R_BURN, ...)	Done
☐	All net labels spelled identically. Key nets: +3V3_SYS, GND, BURN_CTRL, PULSE_IN, SPI_SCK, SPI_MISO, SPI_MOSI, RADIO_CS, RADIO_RST, RADIO_DIO0, FLASH_CS, SPI2_SCK, SPI2_MISO, SPI2_MOSI, I2C_SCL, I2C_SDA, HV_EN, HV_MON, DEPLOY_SENSE, EPS_CHRG_OK, UART_TX, UART_RX	Done
☐	Pin 4 crystal wiring	Fixed
☐	Unused pins have no-connect flags (section 12)	To check
☐	Run ERC in the schematic and review every warning	To do
☐	Confirm in EasyEDA that the Y1 symbol pins 1 and 3 are the crystal, and 2 and 4 are GND	To do
☐	Confirm the flash footprint is 208-mil SOIC-8 (not 150-mil)	To do
☐	Verify the 2x20 stack header part (long-pin stacking type)	To do
☐	Run DRC after layout	To do

16. Open items and changes needed on other boards

> These items are **not yet verified**. Do not treat them as final.

16.1 Open on Stage 3

- ERC has **not been run** yet.
- Layout and copper pours are **not done** yet.
- 2x20 stack header part number is **unverified**.
- Y1 EasyEDA symbol pin mapping and the flash footprint still need to be **confirmed**.
- Crystal capacitor values may need **tuning** on the first board.
- 8 MHz crystal drive level to be **measured** on the first board (this decides R_EXT).
- DEPLOY_SENSE pull-up or pull-down choice depends on the Stage 6 design.
- Stage 2 and Stage 6 designs have **not been seen**, so their header signals are plans to confirm.
- The mode state machine (which mode the satellite is in, and when it switches) is firmware work for later.

16.2 Changes needed on other boards

Board	Change
Stage 4	Add a 100 k pull-down on the burn MOSFET (AO3400A) gate
Stage 4	Remove the duplicate I2C pull-ups
Stage 4	Route CHRG_OK (BQ25713 pin 4) to left header pin 25
Stage 6	RFM96W or RFM98W radio module

Board	Change
Stage 6	100 nF + 10 µF at the radio supply pin
Stage 6	u.FL antenna connector
Stage 6	Matching header pins (section 11)
Stage 6	DEPLOY_SENSE on left header pin 24
Stage 2	33 to 100 Ω series resistor on the PULSE output
Stage 2	100 k pull-down on HV_EN
Stage 2	HV_MON divided down to 3.3 V or less, on left header pin 23

17. Glossary

Term	Plain meaning
Net	One electrical connection: everything wired together shares one net name.
Net label	A name tag in the schematic; wires with the same label are connected even without a drawn line.
Pull-up / pull-down resistor	A resistor to 3.3 V (up) or GND (down) that holds a signal at a known level when nothing drives it.
Decoupling capacitor	A small capacitor right at a chip's power pin that smooths fast supply dips, like a tiny local battery.
Bulk capacitor	A larger capacitor that smooths slower supply dips for the whole chip.
ESR	The small built-in resistance of a capacitor.
Crystal	A small quartz timing part that keeps the chip's clock accurate (not the gamma-ray detector crystal).
SPI	A fast 4-wire link: clock, data out, data in, and a chip-select line to pick which chip is listening.
I2C	A 2-wire shared bus (clock and data) where every chip has its own address.
UART	A simple 2-wire serial link, one wire to send (TX) and one to receive (RX).
SWD	ST's 2-wire programming and debug connection, used with an ST-Link.
ADC	A circuit inside the chip that measures a voltage and turns it into a number.
GPIO	A chip pin used as a plain on/off input or output.
Firmware	The program that runs on the microcontroller.
Bootloader	A small built-in loader program that can install firmware without SWD.
Pour / plane	A large filled area of copper on one layer, connected to one net (for example GND).
Via	A small plated hole that connects copper on different layers.
Footprint	The copper pad pattern on the board that a part is soldered to.
ERC	Electrical Rule Check: the schematic checker for wiring mistakes.
DRC	Design Rule Check: the layout checker for spacing, widths and holes the factory can make.

Term	Plain meaning
LCSC / JLCPCB	LCSC is a parts supplier; JLCPCB is the board maker and assembler that uses LCSC parts.
Basic vs Extended part	At JLCPCB, Basic parts are always loaded on the machines (cheaper); Extended parts must be loaded for your job (extra fee).

18. Sources

All files are in `/workspace/cubesat-research/`.

Document	What it covers	File
ST DS10314	STM32F411xC/xE datasheet (pinout Table 8, alternate functions Table 9, VCAP Table 16)	ds/ds10314.pdf
ST AN4488	Getting started with STM32F4 hardware development (power, decoupling, boot)	ds/an4488.pdf
ST RM0383	STM32F411 reference manual (boot modes, GPIO reset state, LSEMOD)	ds/rm0383.pdf
ST AN2867	Oscillator design guide for ST microcontrollers (crystal choice, load caps, layout)	ds/an2867.pdf
Semtech SX1276/77/78/79	Radio chip datasheet (reset rule section 7.2, SPI speed)	ds/sx1276_77_78_79.pdf
HopeRF RFM95/96/97/98(W)	Radio module datasheet (RFM95W = 868/915 MHz; RFM96W/98W = 433 MHz)	ds/RFM95_96_97_98W.pdf
Winbond W25Q128JV	Flash memory datasheet, plus LCSC copies	ds/flash/
Crystal and capacitor sheets	YXC X32258MOB4SI (C2682775), Epson FC-135R (C94674), C1644, C1662	ds/xtal/
TI BQ25713	Battery charger datasheet (I2C address, CHRG_OK pin)	ds/bq25713.pdf
NXP UM10204, TI SLVA689	I2C bus specification; I2C pull-up calculation	ds/UM10204.pdf, ds/slva689.pdf
Project notes	Stage3-CDH-Wiring-Review.md (sections 3.3, 3.4 and 4 outdated), Stage3-Step1-Parts.md, Stage3-Step7-Crystals.md, Stage3-Step8-Storage.md	cubesat-research/